



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES
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BIOS vs UEFI Comparison

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1. Introduction

BIOS and UEFI are firmware technologies responsible for initializing computer hardware and starting the operating system boot process. BIOS (Basic Input/Output System) is the traditional firmware interface used on older systems, while UEFI (Unified Extensible Firmware Interface) is the modern firmware standard used by most current computers, including the virtual machines used in this project.

2. Comparison Table

Category	BIOS	UEFI
Definition	Traditional firmware interface used to initialize hardware and begin booting	Modern firmware interface designed to replace traditional BIOS
Boot Process	Uses a 16-bit real-mode boot process with limited pre-boot functionality	Uses a more modern boot process with native support for EFI executables and a richer pre-boot environment
Maximum Disk Support	Traditionally limited to disks around 2 TB due to MBR	Supports GPT and disks well beyond 2 TB
Partition Style	Commonly uses MBR (Master	Commonly uses GPT (GUID



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Category	BIOS	UEFI
	Boot Record), limited to 4 primary partitions	Partition Table), supports up to 128 partitions
Security Features	More limited built-in security capabilities	Supports features such as Secure Boot and firmware-level malware protection
Speed	Generally slower during startup due to sequential hardware initialization	Generally faster startup through more efficient initialization
Advantages	Simple, widely understood, compatible with older systems	Modern design, GPT support, Secure Boot, larger disk support, graphical pre-boot interface
Disadvantages	Older architecture with limitations on modern storage and security	More complex than traditional BIOS
Modern Usage	Primarily found on legacy systems	Standard firmware approach on virtually all modern systems



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3. Why UEFI Replaced BIOS

UEFI has largely replaced BIOS because modern computing environments require capabilities that the traditional BIOS was never designed to provide.

Storage support. Traditional BIOS systems commonly rely on the MBR partitioning scheme, which restricts disks to roughly 2 TB and a maximum of four primary partitions. UEFI works natively with GPT, which supports much larger storage devices and a significantly higher number of partitions an important consideration for enterprise servers handling large volumes of data.

Security. UEFI introduces capabilities that BIOS lacks, most notably Secure Boot, which verifies that only trusted, digitally signed boot components are allowed to run during startup. This helps protect against boot-level malware such as rootkits.

Extensibility and speed. UEFI was designed as a more extensible firmware environment, supporting drivers, a mouse-capable graphical pre-boot interface, and faster initialization than the sequential, text-based BIOS process. Instead of relying on the limitations of the traditional BIOS interface, UEFI provides a modern framework for firmware applications and boot managers.

For modern enterprise systems including the Ubuntu Server virtual machine deployed in this project these capabilities make UEFI a better fit than legacy BIOS. BIOS remains relevant mainly for older hardware and specific compatibility scenarios, but UEFI is the preferred firmware approach for current and future systems.



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4. Conclusion

BIOS and UEFI both perform the essential function of preparing hardware and starting the operating system. However, UEFI provides significantly greater support for modern storage capacities, boot-level security, and firmware extensibility. For modern computer systems, including enterprise servers, UEFI is generally the preferred choice because it better supports current hardware and operating system requirements.